Factorising and Expanding Brackets

Fluency

Expand

(a) $3(x+4)$

(b) $5(2x-3)$

(c) $-2(x+6)$

(d) $4(3x-5)$

(e) $-(x-7)$

(f) $x(x+3)$

(g) $2x(x-5)$

(h) $-3x(2x+1)$

Factorise

(a) $4x+12$

(b) $6x-15$

(c) x^2+5x

(d) $3x^2-9x$

(e) $10x+25$

(f) $2x^2+8x$

(g) $9x-6$

(h) $9x^2-6x$

Problem Solving

Expand and simplify

(a) $3(x+2)+2(x-5)$

(b) $4(2x-1)-3(x+2)$

(c) $(x+3)(x+4)$

(d) $(2x-1)(x+5)$

(e) $(x-6)(x-2)$

Factorise fully

Find the highest common factor (HCF) of every term — it can include more than one letter, and a power of a letter.

(a) $18x^5y^2+12x^7$

(b) $8a^3b^2-20a^2b^4$

(c) $15x^3y-9x^2y^2+6xy^3$

(d) $14p^4q^3+21p^2q^5$

(e) $24m^6n^2-16m^4n^5+8m^3n^3$

Solve

Expand the brackets first, then solve for x.

(a) $3(x+5)=27$

(b) $2(3x-4)=5x+1$

(c) $5(x-2)-3(x+1)=7$

A step further

(a) A rectangle has length $(x+5)$ and width $(x-2)$. Write an expression for its area, expanded and simplified.

(b) The length of a rectangle is $2x-3$ and its width is x. The area is $2x^2-3x$. Factorise this area expression to check it matches the length and width given.

Reasoning

(a) Write 98 as $(100 - 2)$. Use expanding to calculate 7×98.

(b) Use factorising to quickly calculate $37 \times 8 + 37 \times 2$.

(c) A fruit stall sells 24 apples at \$2 each and 24 oranges at \$3 each. Use factorising to calculate $24 \times 2 + 24 \times 3$ in one step.

(d) Write 203 as $(200 + 3)$. Use expanding to calculate 6×203.

(e) Write 104 as $(100 + 4)$. Use expanding to calculate 9×104.

(f) Use factorising to calculate $85 \times 12 - 85 \times 2$.

Enrichment

Factorising Quadratics

A quadratic $x^2 + bx + c$ (where x^2 has no coefficient) factorises as $(x + p)(x + q)$, where p and q are two numbers that multiply *to give c and* add *to give b. If the quadratic has a coefficient in front of the x^2, like $ax^2 + bx + c$, find two numbers that multiply to give $a \times c$ and add to give b, use them to split the middle term into two terms, then factorise by grouping. A difference of two squares, $a^2 - b^2$, always factorises as $(a - b)(a + b)$.*

Worked example (no coefficient): Factorise $x^2 + 5x + 6$. Two numbers that multiply to 6 and add to 5 are 2 and 3, so $x^2 + 5x + 6 = (x + 2)(x + 3)$.

Worked example (with a coefficient): Factorise $2x^2 + 7x + 3$. Here $a \times c = 2 \times 3 = 6$, and we need two numbers that multiply to 6 and add to 7: these are 1 and 6. Split the middle term and group:

$$2x^2 + 7x + 3 = 2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3).$$

(a) $x^2 + 7x + 10$

(b) $x^2 - 3x - 18$

(c) $x^2 - 16$

(d) $4x^2 - 9$

(e) $2x^2 + x - 6$

Inequalities

Inequalities behave like equations, except multiplying or dividing by a negative number reverses the inequality sign.

Worked example: Solve $2(x - 3) < 10$.

$$2(x - 3) < 10 \;\Rightarrow\; 2x - 6 < 10 \;\Rightarrow\; 2x < 16 \;\Rightarrow\; x < 8.$$

(Expand the bracket, add 6 to both sides, then divide both sides by 2.)

Worked example (dividing by a negative): Solve $-2x + 5 > 1$.

$$-2x + 5 > 1 \;\Rightarrow\; -2x > -4 \;\Rightarrow\; x < 2.$$

(Subtract 5 from both sides, then divide both sides by -2 — since we divided by a negative number, the $>$ sign flips to $<$.)

(a) Solve: $3(x + 2) > 15$

(b) Solve: $2(x - 5) \leq 4(x + 1)$

(c) Solve: $-3(x + 1) < 2(x - 4)$

(d) Solve the double inequality: $2 < 3(x - 1) \leq 11$

(e) A taxi charges a \$4 call-out fee plus \$3 per kilometre travelled. Write an inequality for the total cost of a trip of k kilometres, then solve it to find how far you could travel and keep the cost under \$40.
